

Balanced Multi-View Clustering

Zhenglai Li¹, Jun Wang², Chang Tang³, Senior Member, IEEE, Xinzhong Zhu⁴, Member, IEEE, Wei Zhang⁵, and Xinwang Liu⁶, Senior Member, IEEE

Abstract—Multi-view clustering (MvC) aims to integrate information from different views to enhance the capability of the model in capturing the underlying data structures. The widely used joint training paradigm in MvC potentially does not fully leverage the multi-view information, due to the imbalanced and under-optimized view-specific features caused by the uniform learning objective for all views. For instance, particular views with more discriminative information could dominate the learning process in the joint training paradigm, leading to other views being under-optimized. To alleviate this issue, we first analyze the imbalanced phenomenon in the joint-training paradigm of multi-view clustering from the perspective of gradient descent for each view-specific feature extractor. Then, we propose a novel balanced multi-view clustering (BMvC) method, which introduces a view-specific contrastive regularization (VCR) to modulate the optimization of each view. Concretely, VCR preserves the sample similarities captured from the joint features and view-specific ones into the clustering distributions corresponding to view-specific features to enhance the learning process of view-specific feature extractors. Additionally, an analysis is provided to illustrate that VCR adaptively modulates the magnitudes of gradients for updating the parameters of view-specific feature extractors to achieve a balanced multi-view learning procedure. In such a manner, BMvC achieves a better

trade-off between the exploitation of view-specific patterns and the exploration of view-invariance patterns to fully learn the multi-view information for the clustering task. Finally, a set of experiments are conducted to verify the superiority of the proposed method compared with state-of-the-art approaches both on eight benchmark MvC datasets and two spatially resolved transcriptomics datasets.

Index Terms—Balanced learning, contrastive learning, multi-view clustering, multi-view learning, view cooperation.

I. INTRODUCTION

IN REAL-WORLD applications, data are often collected from multiple sensors or depicted by diverse feature descriptors. For example, the information captured from different sensors (e.g., cameras, LiDARs, Radars) is fused to achieve comprehensive scene understanding for autonomous vehicles [1]. The text, visual, and audio are jointly utilized in the video process [2]. The traditional shallow features descriptors, e.g., histogram of oriented gradients (HOG) [3], and deep features descriptors, e.g., transformer [4], are both leveraged to depict the objects in some computer vision tasks [5]. Accordingly, multi-view learning methods, which explore the complementary information from diverse views to improve the model performance, have witnessed great progress in these years [6], [7], [8].

Clustering algorithms aim to group the data points into their respective clusters without labels. As a useful unsupervised learning tool to reveal the underlying semantic structure of data for capturing meaningful information, clustering algorithms have been widely studied and employed in various fields, such as social network [9], image segmentation [10], and single-cell RNA-seq data clustering [11]. As one of the kinds of multi-view learning approaches, multi-view clustering expects to learn a shared and comprehensive clustering presentation by utilizing the complementary information from diverse views, to boost the clustering performance, which has received much attention recently [12].

Existing multi-view clustering approaches usually employ a joint training paradigm to optimize the clustering models as shown in Fig. 1(a), which integrates the view-specific features into a shared representation under a uniform joint training objective, i.e., reconstructing the features (Rec Loss) or clustering (Clu Loss) by the shared representation, to extract the comprehensive information for the clustering task. For example, Xie et al. [13] designed a joint learning framework, which learns diverse feature representations for images via different neural networks and fuses the information to obtain the clustering results. The work in [14], designs a framework to investigate the effects of self-supervision and contrastive alignment in

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Zhenglai Li is with the Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China (e-mail: zhenglai.li@siat.ac.cn Member).

Jun Wang and Xinwang Liu are with the school of computer, National University of Defense Technology, Changsha 410073, China (e-mail: wang_jun@nudt.edu.cn; xinwangliu@nudt.edu.cn).

Chang Tang is with the School of Software Engineering, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: tangchang@hust.edu.cn).

Xinzhong Zhu is with the Zhejiang Key Laboratory of Intelligent Education Technology and Application, Zhejiang Normal University, Jinhua 321004, China, and also with the School of Computer Science and Technology, Zhejiang Normal University, Jinhua 321004, China (e-mail: zxz@zjnu.edu.cn).

Wei Zhang is with the Key Laboratory of Computing Power Network and Information Security, Ministry of Education, Shandong Computer Science Center (National Supercomputer Center in Jinan), Qilu University of Technology (Shandong Academy of Sciences), Jinan 250000, China, and also with the Shandong Provincial Key Laboratory of Computing Power Internet and Service Computing, Shandong Fundamental Research Center for Computer Science, Jinan 250000, China (e-mail: wzhang@qlu.edu.cn).

The demo code of this work is publicly available at <https://github.com/guanyuezheng/BMvC>.

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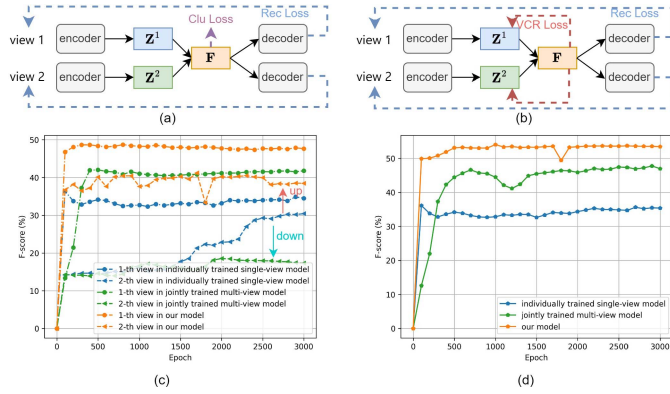


Fig. 1. Illustration of the imbalanced multi-view clustering issue in a two-view case. (a) The joint training multi-view clustering paradigm. (b) Our proposed balanced multi-view clustering approach. (c) The clustering performance of view-specific features extracted by diverse models. (d) The clustering performance of the joint representation obtained by different models.

deep multi-view clustering. In recent studies, some researchers claimed that the clustering performance of the multi-view model may degrade when the number of views increases [15], [16]. The main reason is that the particular views with noisy information could be not only useless but also even detrimental for the clustering task [15]. This discrepancy makes it challenging for the multi-view clustering model to effectively learn from all views simultaneously under a uniform joint training objective. To alleviate this issue, previous methods usually select some informative views for clustering [16], [17], but this also leads to insufficient utilization of the complementary information among all views.

Beyond the clustering performance degradation caused by diverse view qualities, we observe that even when the multi-view clustering models perform better than their single-view counterparts, they still fail to fully explore the potential of multiple views. As shown in Fig. 1, a set of experiments are conducted on the nuswide dataset [18] to evaluate the performance of diverse multi-view clustering settings. From the results, we note that the joint training multi-view clustering models outperform single-view ones, but the clustering performance of view-specific features within the jointly-trained multi-view clustering models performs similarly even worse than the performance of view-specific features captured from solely trained single-view clustering models. For example, in Fig. 1(c), 1-th view features in the jointly-trained multi-view clustering model show a clear clustering performance drop, compared with that in the single-view model. Such observations indicate that the view-specific feature extractors are under-optimized with an imbalanced degree in the joint training multi-view clustering model. The main reason is that the multi-view datasets often contain some views with more discriminative information, which tend to be favored and dominant in the training procedure, consequently suppressing the learning process of other views. Such a view preference within the datasets causes the observed imbalanced multi-view clustering issue among different views.

Although many solid MVC methods have been proposed [6], [12], the efforts to tackle the imbalance multi-view clustering issue are still limited. To this end, we propose BMvC (short for balanced multi-view clustering) as illustrated in Fig. 1(b).

a novel solution that introduces a view-specific contrastive regularization (VCR) to modulate the optimization of each view. Concretely, VCR constrains the clustering distributions of view-specific features to preserve the sample similarities captured from the joint features and view-specific ones. As a result, the VCR provides an additional gradient for updating view-specific feature extractors so that a better trade-off between the exploitation of view-specific patterns and exploration of view-invariance patterns can be achieved to fully learn the multi-view information for the clustering task. As shown in Fig. 1(c), the clustering performance of view-specific features extracted by our proposed method outperforms better than that of joint training and single-view training models. Finally, the proposed method achieves better results compared with the joint training model as given in Fig. 1(d). The contributions are summarized as follows,

- We observe and analyze the phenomenon of imbalanced learning in multi-view clustering from the gradient descent perspective. In the joint training paradigm, view-specific encoders are under-optimized, leading to certain views being worse optimized than others during the training process.
- We propose BMvC to achieve a better trade-off between the exploitation of view-specific patterns and the exploration of view-invariance patterns to fully learn the multi-view information for the clustering task.
- We introduce VCR to exploit the sample similarities captured from the joint features and view-specific ones to regularize the clustering distributions of view-specific features being structural, so that the view-specific encoders are learned balanced. A theoretical analysis is formulated to demonstrate that VCR adaptively modulates the gradients for updating the parameters of view-specific encoders to achieve balanced learning processing.
- We perform extensive experiments on eight benchmark multi-view datasets to verify the effectiveness of the proposed method. Additionally, our BMvC method is extended to apply to the spatially resolved transcriptomics data clustering task to demonstrate the efficacy of the proposed method.

The rest of this paper is organized as follows. Section II gives a brief review of the most related work. In Section III, we present the details of the proposed BMvC method. Section IV provides a series of experimental results and discussions. In Section V, we provide a conclusion of this paper. In this paper, several experiments are included in the Appendix (online available) due to limited space.

II. RELATED WORK

A. Multi-View Clustering

Multi-view clustering, which aggregates comprehensive information from various views, has attracted considerable attention due to the increasing amount of multi-view data. Current methods can be classified into three primary categories based on their approach to generating clustering results: matrix factorization-based methods [19], [20], [21], graph-based methods [22], [23], [24], [25], [26], [27], and deep learning-based methods [13], [15], [17], [28], [29].

Matrix factorization-based methods: They aim to derive a shared feature representation from multiple views through matrix factorization techniques. For instance, Liu et al. [30] developed a shared factorization that provides compatible solutions across diverse views to achieve clustering outcomes. Building upon the deep matrix factorization methods [31], Zhao et al. [21] hierarchically decomposed the multi-view representations into a shared latent feature space, progressively learning complementary information from the various views. They employed the Laplacian regularization to maintain the locality properties of the data within this latent space. Yang et al. [20] introduced a tri-factorization-based non-negative matrix factorization approach to decompose multi-view data into a uniform distribution, thereby enhancing the separability of the learned consensus representation. To derive clustering results from multiple graph embeddings, the method in [32] introduces view-specific rotation matrices that project view-specific embedding matrices into a view-shared cluster indicator matrix for clustering. Furthermore, Wang et al. [33] proposed anchor graphs to reduce computational complexity and designed an effective one-step approach to obtain clustering results from the anchor graph embedding. Zhang et al. [34] introduced an adaptive cluster-wise anchor learning method that enforces an explicit cluster structure among latent anchors through a consensus cluster indicator matrix, enhancing both between-cluster diversity and within-cluster consistency to improve subspace representation discrimination. To address the challenges posed by the incomplete multi-view data, Wen et al. [35] implemented an adaptive feature weighting constraint aimed at mitigating the effects of redundant and noisy features during the matrix factorization process. Additionally, they designed a graph-embedded consensus representation learning term to preserve the structural information inherent in incomplete multi-view data.

Graph-based methods: Graphs are widely used data structures for representing the relationships among different samples. Graph-based multi-view clustering methods integrate graph similarities learned from various views, ultimately deriving clustering results through spectral clustering techniques. Self-representation [36] and adaptive neighbor graph learning [37] are two prevalent approaches for constructing high-quality similarity graphs in prior works. For instance, Zhang et al. [22] jointly learned both latent and subspace representations, employing neural networks to enhance the latent representation learning process and improve model generalization. Nie et al. [38] extended adaptive neighbor graph learning to a multi-view setting and captured a Laplacian rank-constrained consensus graph from multiple views for clustering. Tang et al. [39] unified k -means and spectral clustering to leverage information from both graphs and embedding matrices, thereby enhancing clustering performance. Anchor graph learning is an effective approach for reducing the computational complexity of models. Recently, several multi-view clustering methods based on anchor graphs have been proposed. For instance, in [40], a joint learning framework is developed that simultaneously learns the graph embedding matrix and clustering indicators, effectively unifying spectral embedding and spectral rotation methods. Kang et al. [23] designed an efficient anchor graph learning approach

based on anchor-based subspace learning. Fang et al. [41] incorporated anchor-based subspace learning to derive view-specific bipartite graphs and fused them into a view-consensus bipartite graph with adaptive weight learning and a Laplacian rank constraint.

Deep learning-based methods: They leverage the powerful representation capabilities of deep neural networks to derive consensus clustering results from multi-view data. For example, Wang et al. [42] proposed a method that uses deep canonically correlated autoencoders to learn representations from multiple unlabeled views for clustering. Wang et al. [29] utilized a consistent generative adversarial network to capture a shared representation from incomplete multi-view data for clustering. Xie et al. [13] developed a joint learning framework that employs different neural networks to extract features for comprehensively depicting each image. This framework enables simultaneous feature embedding, multi-view information fusion, and data clustering to achieve joint training. Xu et al. [28] introduced a collaborative training scheme to capture complementary information from diverse views, facilitating the joint learning of feature representations and cluster assignments. In recent years, self-supervised learning with contrastive loss has made significant strides and received rapid developments across various fields [43], [44]. The core principle of contrastive learning is to push samples away from negative anchors while pulling them closer to positive anchors, thereby creating discriminative representations for downstream tasks [45]. Over the past few years, various contrastive learning approaches for both unimodal and multimodal representation learning have emerged, including MoCo [46], SimCLR [47], and MoCo-MM [48]. For a comprehensive overview of additional methods, we refer to the survey in [49]. Drawing inspiration from the robust feature learning capabilities of self-supervised learning, contrastive loss has been extensively applied in multi-view clustering [50], [51]. For example, Xu et al. [52] aligned multi-view information from both high-level semantics and low-level features through contrastive learning, effectively capturing the common semantics for clustering. Lin et al. [53] integrated feature representation learning and missing sample recovery into a unified framework, using contrastive learning to capture informative and consistent representations from different views. Additionally, the work in [14] designed a framework to explore the effectiveness of self-supervision and contrastive alignment in the multi-view clustering task. In [54], the instance-wise contrastive loss is utilized to maximize the consistency between view-consensus and global representations. Additionally, a self-weighting method that measures pairwise representation discrepancies is employed to enhance useful views while diminishing the influence of unreliable ones. In contrast, SCM [55] first fuses multi-view information at the data level and simulates data noise and unavailability across multiple views to construct two augmentations for contrastive learning. Previous multi-view clustering methods primarily utilize cross-view instance contrastive learning and cross-view cluster contrastive learning to align instances and clusters across multiple views for feature learning. A common representation is then constructed by merging multi-view features for clustering. The proposed method introduces a view-specific contrastive

regularization term that measures the alignment between graph topologies and cluster distributions. This balances the multi-view learning process, resulting in a fundamentally different loss formulation and design compared to previous contrastive learning-based multi-view clustering methods.

Matrix factorization-based and graph-based multi-view clustering methods usually utilize a shallow method to capture the underlying cluster structures from multi-view data. deep learning based multi-view clustering methods designed based on collaborative training [17], contrastive learning [51], and weighted multi-view feature fusion [54], achieving considerable clustering performance. The core idea of these previous multi-view methods is to seek a shared representation among multiple views, allowing consistent and complementary information from various views to be encoded into a view-shared representation for clustering. However, the issue of imbalanced learning in multi-view clustering, arising from the diverse qualities of views, has been rarely addressed. To this end, we propose a balanced multi-view clustering method aimed at enhancing cooperation among views.

B. Imbalanced Learning

Multi-view and multi-modal learning are two closely intertwined concepts. In the context of multi-modal fusion, studies have shown that joint-training supervised multi-modal learning often encounters modality competition [56], [57]. This issue arises when different modalities are optimized synchronously under a unified objective. During training, modalities with more discriminative information tend to dominate the learning process, converging more quickly than others. As a result, the other modalities struggle to update their learning parameters fully. Consequently, the under-optimized unimodal feature extractors prevent the full exploitation of modality-specific information, generating a bottleneck in boosting the performance of multi-modal learning.

Recently, several methods have emerged to address this issue [58], [59], [60], [61]. For instance, Du et al. [58] employed a knowledge distillation scheme for uni-modal distillation to enhance multimodal model performance. The works in [59], [60] utilize the adaptive gradient modulation to suppress the learning process of the dominant modality and facilitate the training of other modalities to achieve more balanced multi-modal learning. Fan et al. [61] leveraged a prototypical cross-entropy loss to accelerate the learning of weaker modalities. Additionally, the study in [62] introduced a multi-loss objective to dynamically adjust the learning process for each modality, refining the balancing mechanism. However, these previous methods primarily address the imbalanced learning problem in supervised tasks, making them infeasible to be directly applied for the unsupervised multi-view learning task.

III. THE PROPOSED METHOD

Problem Definition: For convenience, the multi-view data is presented as $\{\mathbf{X}^r \in \mathbb{R}^{N \times D^r}\}_{r=1}^M$, where N and M are the number of samples and views. D^r is the feature dimensions of r -th view. The goal of multi-view clustering is to formulate a

projection to capture the underlying data structures, i.e., $\{\mathbf{X}^r \in \mathbb{R}^{N \times D^r}\}_{r=1}^M \rightarrow \mathbf{Y}$, where $\mathbf{Y}$ denotes the clustering results.

In this section, we first analyze the imbalanced learning phenomenon in multi-view clustering. Then, we introduce the proposed balanced multi-view clustering framework. Finally, the analysis related to the proposed method is given.

A. Imbalanced Multi-View Clustering

In this part, we analyze the imbalanced phenomenon in the joint training paradigm of previous multi-view clustering methods. The view-specific encoder $\mathcal{E}^r(\boldsymbol{\eta}^r, \mathbf{X}^r)$ is utilized to extract r -th view-specific latent features $\mathbf{Z}^r$ from $\mathbf{X}^r$ as,

$$\mathbf{Z}^r = \mathcal{E}^r(\boldsymbol{\eta}^r, \mathbf{X}^r), r = 1, 2, \dots, M, \quad (1)$$

where $\boldsymbol{\eta}^r$ is the parameters of r -th view-specific encoder. After the view-specific feature extraction, their features are usually fused by a feature concatenation operation, and passed through a linear layer to obtain the joint representation $\mathbf{F}$, as follows,

$$\mathbf{F} = \mathcal{F}(\boldsymbol{\delta}, [\mathbf{Z}^1, \dots, \mathbf{Z}^M]), \quad (2)$$

where $\mathcal{F}(\boldsymbol{\delta}, [\mathbf{Z}^1, \dots, \mathbf{Z}^M])$ with its parameters $\boldsymbol{\delta}$ represents the fusion module. $[\cdot]$ is a feature concatenation operation. By decomposing $\boldsymbol{\delta}$ as $\boldsymbol{\delta} = [\boldsymbol{\delta}^1; \dots; \boldsymbol{\delta}^M]$, (2) can be rewritten as,

$$\mathbf{F} = \sum_{s=1}^M \boldsymbol{\delta}^s \cdot \mathbf{Z}^s. \quad (3)$$

Finally, the joint representation $\mathbf{F}$ passes to the view-specific decoder $\mathcal{D}^r(\boldsymbol{\theta}^r, \mathbf{Z}^r)$ to reconstruct the view-specific features. The objective of a multi-view clustering network is usually to minimize the reconstruction loss between the original and reconstructed features, and additionally constrains the joint representation $\mathbf{F}$ with a clustering loss,

$$\begin{aligned} \mathcal{L} &= \sum_{r=1}^M \ell(\hat{\mathbf{X}}^r, \mathbf{X}^r) + \beta \psi(\mathbf{F}) \\ &= \sum_{r=1}^M \ell \left(\mathcal{D}^r \left(\sum_{s=1}^M \boldsymbol{\delta}^s \cdot \mathcal{E}^s(\boldsymbol{\eta}^s, \mathbf{X}^s) \right), \mathbf{X}^r \right) \\ &\quad + \beta \psi \left(\sum_{s=1}^M \boldsymbol{\delta}^s \cdot \mathcal{E}^s(\boldsymbol{\eta}^s, \mathbf{X}^s) \right), \end{aligned} \quad (4)$$

where β is a hyper-parameter to balance reconstruction loss $\ell(\cdot)$ and clustering loss $\psi(\cdot)$. During the optimization procedure, the parameters $\boldsymbol{\eta}^s$ in $\mathcal{E}^s(\boldsymbol{\eta}^s, \mathbf{X}^s)$ are updated as,

$$\boldsymbol{\eta}^{s,t+1} = \boldsymbol{\eta}^{s,t} - \gamma \nabla_{\boldsymbol{\eta}^s, t} \mathcal{L} = \boldsymbol{\eta}^{s,t} - \gamma (\Delta),$$

$$\begin{aligned} \text{where } \Delta &= \left(\sum_{r=1}^M \frac{\partial \ell(\mathcal{D}^r(\mathbf{F}), \mathbf{X}^r)}{\partial \mathcal{D}^r(\mathbf{F})} \frac{\partial \mathcal{D}^r(\mathbf{F})}{\partial \mathbf{F}} + \beta \frac{\partial \psi(\mathbf{F})}{\partial \mathbf{F}} \right) \\ &\quad \frac{\partial \boldsymbol{\delta}^s \cdot \mathcal{E}^s(\boldsymbol{\eta}^s, \mathbf{X}^s)}{\partial \boldsymbol{\eta}^s, t}. \end{aligned} \quad (5)$$

In (5), $\sum_{r=1}^M \frac{\partial \ell(\mathcal{D}^r(\mathbf{F}), \mathbf{X}^r)}{\partial \mathcal{D}^r(\mathbf{F})} \frac{\partial \mathcal{D}^r(\mathbf{F})}{\partial \mathbf{F}} + \beta \frac{\partial \psi(\mathbf{F})}{\partial \mathbf{F}}$ is a shared term in gradient descent optimization procedure for parameters in all

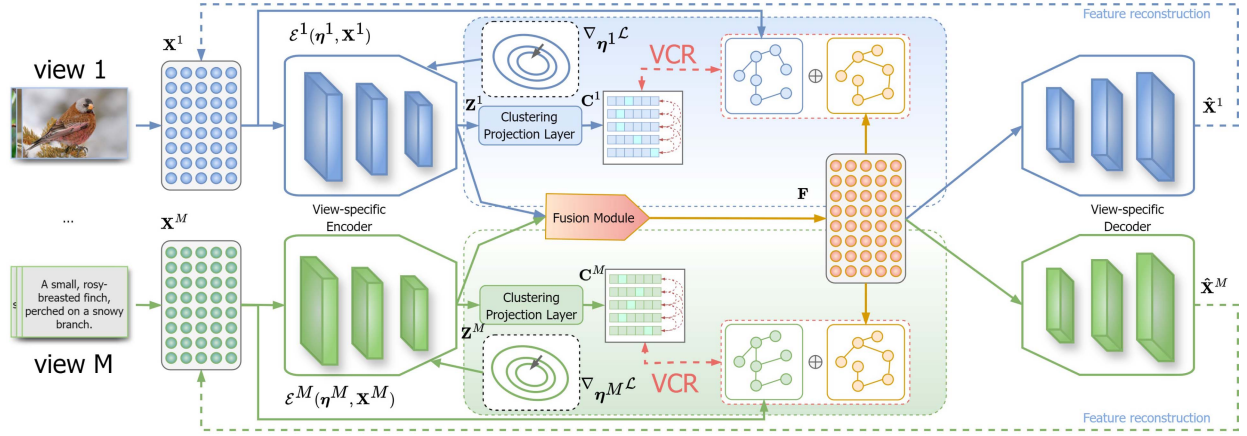


Fig. 2. Illustration of balanced multi-view clustering (BMvC). The multi-view data are first processed through their encoders to extract view-specific latent features. Next, a fusion module aggregates the representations into a unified joint one, which is then used to reconstruct the multi-view data via view-specific decoders. Finally, the view-specific contrastive regularization (VCR) and feature reconstruction loss are employed to guide the optimization of the model.

view-specific encoders. Thus, we can obtain the following observations: 1) The parameters update for encoders corresponding to all views will be stuck simultaneously, when the gradient for the shared term varies small determined by the joint representation $\mathbf{F}$; 2) If certain view, such as the r -th view, consists of more discriminative information than other views, it would dominate the multi-view fusion process for the joint representation $\mathbf{F}$ and the gradient $\sum_{r=1}^M \frac{\partial \ell(\mathcal{D}^r(\mathbf{F}), \mathbf{X}^r)}{\partial \mathcal{D}^r(\mathbf{F})} \frac{\partial \mathcal{D}^r(\mathbf{F})}{\partial \mathbf{F}} + \beta \frac{\partial \psi(\mathbf{F})}{\partial \mathbf{F}}$. Even if other views are under-optimized and their underlying clustering structures are not fully aggregated in the joint representation $\mathbf{F}$, the information from the dominated view can still reconstruct the features. To address this issue, we intend to find a better trade-off between the exploitation of view-specific patterns and the exploration of view-invariance patterns to fully learn the multi-view information for the clustering task.

B. Balanced Multi-View Clustering

Our proposed balanced multi-view clustering (BMvC) method follows the same network architecture as the joint training paradigm approaches depicted in Section III-A. As shown in Fig. 2, the multi-view data $\{\mathbf{X}^r \in \mathbb{R}^{N \times D^r}\}_{r=1}^M$ are processed by their view-specific encoders $\{\mathcal{E}^r(\eta^r, \mathbf{X}^r)\}_{r=1}^M$ to extract the view-specific latent features $\{\mathbf{Z}^r\}_{r=1}^M$. Then, the fusion module $\mathbf{F} = \sum_{s=1}^M \delta^s \cdot \mathbf{Z}^s$ is introduced to aggregate the multi-view representations into a joint one, which is utilized to reconstruct the multi-view data by view-specific decoders $\{\mathcal{D}^r(\theta^r, \mathbf{F})\}_{r=1}^M$. To boost the exploitation of view-specific patterns and the exploration of view-invariance patterns, we introduce a view-specific contrastive regularization (VCR), which aims to maximize the similarities of view-specific clustering indicators within their neighborhood ones based on the similarities extracted from the view-specific features and the aggregated joint ones. In such a paradigm, the feature extraction capabilities of the view-specific encoders are enhanced since the VCR introduces an extra gradient to optimize the parameters of view-specific encoders, which is analyzed in Section III-B2.

Thus, the total objective function of the proposed method is,

$$\mathcal{L} = \sum_{r=1}^M \ell(\hat{\mathbf{X}}^r, \mathbf{X}^r) + \sum_{r=1}^M \lambda \zeta(\mathbf{F}, \mathbf{Z}^r) \quad (6)$$

where λ is trade-off hyper-parameter to balance reconstruction loss $\ell(\cdot)$ and view-specific contrastive regularization loss $\zeta(\cdot)$. Finally, the clustering results is obtained by conducting K-means clustering on the learned joint representation $\mathbf{F}$.

1) *View-Specific Contrastive Regularization*: Preserving the similarity from the original feature space is a useful scheme to enhance the representation capability of the latent features [63], [64], [65]. Inspired by this, we impose constraints on the view-specific feature representations using structural information to improve their representation capabilities, ensuring a more balanced learning between weak and strong views. To achieve this, we introduce a clustering projection layer that captures the view-specific cluster distributions, aligning them with the structural information derived from both view-specific graphs and a view-common graph. The alignment process is facilitated by a carefully designed view-specific contrastive regularization loss.

Clustering Projection Layer: We project the view-specific representations $\{\mathbf{Z}^r\}_{r=1}^M$ into the clustering space by a linear layer. Then, we employ the QR decomposition [66] to ensure the orthogonal property of clustering indicators in the clustering space. As a result, the view-specific clustering indicator matrices $\{\mathbf{C}^s\}_{s=1}^M$, ensuring the orthogonality property in the clustering space, are obtained to reveal the view-specific cluster distributions.

Similarity Graph Construction: Similarity graphs are widely used to depict the relationships among different samples. In this part, we employ the similarities to enhance the feature discriminative capability of the view-specific features. The CAN [37] is a useful adaptive neighbor graph learning manner, which measures the similarities among different samples as the

link probabilities,

$$\begin{aligned} \min_{\mathbf{P}} \quad & \sum_{i,j=1}^N (\|\mathbf{q}_i - \mathbf{q}_j\|_2^2) p_{ij} + \alpha p_{ij}^2 \\ \text{s.t.} \quad & \sum_{j=1}^n p_{ij} = 1, p_{ij} > 0, \end{aligned} \quad (7)$$

where α is a hyper-parameter. p_{ij} denotes the similarity between sample $\mathbf{q}_i$ and $\mathbf{q}_j$. According to the Lagrangian Augmentation and the Karush-Kuhn-Tucker condition, the closed-form solution of (7) can be obtained as [37],

$$p_{ij} = \frac{\hat{d}_{i,k+1} - \hat{d}_{ij}}{k\hat{d}_{i,k+1} - \sum_{j=1}^k \hat{d}_{ij}} \quad (8)$$

where k is the number of nearest neighbors, $\hat{d}_{ij}$ represents the ij -th element in matrix $\hat{\mathbf{D}}$, which is formulated by sorting matrix $\mathbf{D}$, with its ij -th element $d_{ij} = \|\mathbf{q}_i - \mathbf{q}_j\|_2^2$, from small to large.

Based on the similarity graph construction manner in (8), we can formulate the view-specific graphs $\{\mathbf{G}^s\}_{s=1}^M$ and joint graph $\bar{\mathbf{G}}$ from the view-specific original features $\{\mathbf{X}^s\}_{s=1}^M$ and joint features $\mathbf{F}$. The joint graph $\bar{\mathbf{G}}$ depicts the sample relationships with the multi-view information, while the view-specific graphs $\{\mathbf{G}^s\}_{s=1}^M$ measure the correlations among samples only with their view-specific information. To simultaneously keep the view-specific and view-joint knowledge, the view-specific graphs $\{\mathbf{G}^s\}_{s=1}^M$ and joint graph $\bar{\mathbf{G}}$ are fused as follows,

$$\hat{\mathbf{G}}^s = \frac{\mathbf{G}^s + \bar{\mathbf{G}}}{2}, \quad s = 1, \dots, M, \quad (9)$$

where $\hat{\mathbf{G}}^s$ denotes the enhanced s -th view-specific graph.

Contrastive Loss: The contrastive learning approaches push the samples far away from the negative anchors, while pulling in the samples with positive anchors [47], [67]. The view-specific similarity graphs $\{\hat{\mathbf{G}}^s\}_{s=1}^M$ provide the positive and negative pairs for contrastive learning corresponding to the view-specific clustering indicators $\{\mathbf{C}^s\}_{s=1}^M$. To keep the structural information into the view-specific clustering indicators, we first compute the sample correlations via the cosine similarity as,

$$a_{ij}^s = \frac{\mathbf{c}_i^s \mathbf{c}_j^{s\top}}{\|\mathbf{c}_i^s\| \|\mathbf{c}_j^s\|}, \quad (10)$$

where $a_{ij}^s \in [0, 1]$ is the correlation between $\mathbf{c}_i^s$ and $\mathbf{c}_j^s$. Ideally, if the $\mathbf{c}_i^s$ and $\mathbf{c}_j^s$ are the positive pair, a_{ij}^s is closed to 1. And if $\mathbf{c}_i^s$ and $\mathbf{c}_h^s$ are the negative pair, a_{ih}^s is closed to 0. Thus, we can formulate the following objective function as,

$$\min_{\mathbf{C}^s} = \sum_{i=1}^N \left(\sum_{j \in N^+} (1 - a_{ij}^s)^2 + \sum_{h \in N^-} (a_{ih}^s)^2 \right), \quad (11)$$

where N^+ and N^- denote the positive and negative sets, respectively. Considering the view-specific similarity graphs $\{\hat{\mathbf{G}}^s\}_{s=1}^M$ measure the data relationships in the range $[0, 1]$ rather than the binary value, i.e., 0 or 1, we reformulate the (11) with a weighted

Algorithm 1: Balanced Multi-View Clustering.

Input: BMvC Model, dataset $\{\mathbf{X}^r\}_{r=1}^M$, number of clusters, number of nearest neighbors, hyper-parameter λ , and maximum training epoch T ;

Output: Clustering results $\mathbf{Y}$;

```

1 for  $s = 1, \dots, T$  do
2   Extract view-specific features  $\{\mathbf{Z}^r\}_{r=1}^M$  via Eq. (1);
3   Fuse view-specific features to obtain view-shared
   feature representation  $\mathbf{F}$  via Eq. (2);
4   Construct view-specific graphs  $\{\mathbf{G}^r\}_{r=1}^M$  and
   view-shared graph  $\bar{\mathbf{G}}$  via adaptive neighbor
   graph learning manner in Eq. (7);
5   Pass through the clustering projection layer to
   obtain view-specific clustering indicator
   matrices  $\{\mathbf{C}^r\}_{r=1}^M$ ;
6   Obtain view-specific contrastive loss  $\zeta$  via Eq.
   (12);
7   Perform feature reconstruction and compute
   feature reconstruction loss  $\ell$ ;
8   Do back-propagation and update parameters of
   the BMvC Model;
9 end
10 Perform  $k$ -means clustering algorithm on the
   learned view-shared representation  $\mathbf{F}$  to obtain
   clustering results  $\mathbf{Y}$ .

```

manner as,

$$\zeta = \frac{\sum_{i,j=1}^N (\hat{g}_{ij}^s (1 - a_{ij}^s))^2}{\sum_{i,j=1}^N \hat{g}_{ij}^s} + \frac{\sum_{i,j=1}^N ((1 - \hat{g}_{ij}^s) (a_{ij}^s))^2}{\sum_{i,j=1}^N (1 - \hat{g}_{ij}^s)}. \quad (12)$$

In (12), the first term is used to constrain the samples more closed to each other when they are in the same neighborhoods, while the second term encourages pushing the samples away from others as they are in distinct neighborhoods. In such a manner, the clustering distributions of view-specific features are well learned under the guidance of view-specific graphs and a joint graph. Thus, the view-specific encoders can be further optimized to enhance their feature extraction capabilities.

To clearly outline the entire procedure of the proposed method, we summarize the algorithm in Algorithm 1.

2) The Resultant Gradient Analysis: During the optimization procedure, the gradient from the view-specific contrastive regularization for updating the parameters $\boldsymbol{\eta}^s$ in $\mathcal{E}^s(\boldsymbol{\eta}^s, \mathbf{X}^s)$ is represented as,

$$\begin{aligned} \nabla_{\boldsymbol{\eta}^s} \sum_{r=1}^M \zeta \left(\sum_{s=1}^M \boldsymbol{\delta}^s \cdot \mathcal{E}^s(\boldsymbol{\eta}^s, \mathbf{X}^s), \mathcal{E}^r(\boldsymbol{\eta}^r, \mathbf{X}^r) \right) \\ = \frac{\partial \zeta(\mathbf{F}, \mathbf{Z}^s)}{\partial \mathbf{F}} \frac{\partial \boldsymbol{\delta}^s \cdot \mathcal{E}^s(\boldsymbol{\eta}^{s,t}, \mathbf{X}^s)}{\partial \boldsymbol{\eta}^{s,t}} \\ + \frac{\partial \zeta(\mathbf{F}, \mathbf{Z}^s)}{\partial \mathbf{Z}^s} \frac{\partial \mathcal{E}^s(\boldsymbol{\eta}^{s,t}, \mathbf{X}^s)}{\partial \boldsymbol{\eta}^{s,t}}. \end{aligned} \quad (13)$$

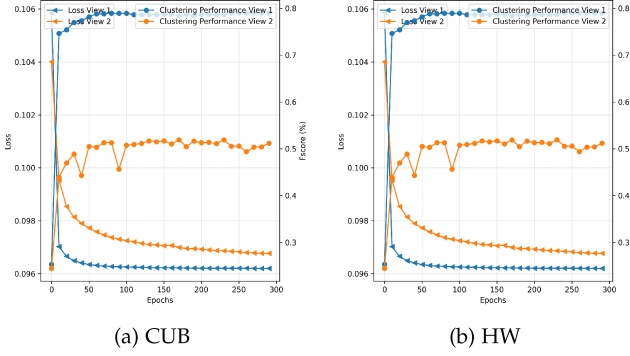


Fig. 3. The clustering performance measured by Fscore and values of $\zeta(\mathbf{F}, \mathbf{Z}^s)$ vary with the number of training epochs. (a) CUB. (b) HW.

Thus, the parameters η^s in $\mathcal{E}^s(\eta^s, \mathbf{X}^s)$ are updated as,

$$\eta^{s,t+1} = \eta^{s,t} - \gamma(\Delta_1 + \lambda\Delta_2),$$

$$\text{where } \Delta_1 = \left(\sum_{r=1}^M \frac{\partial \ell(\mathcal{D}^r(\mathbf{F}), \mathbf{X}^r)}{\partial \mathcal{D}^r(\mathbf{F})} \frac{\partial \mathcal{D}^r(\mathbf{F})}{\partial \mathbf{F}} \right) \frac{\partial \delta^s \cdot \mathcal{E}^s(\eta^{s,t}, \mathbf{X}^s)}{\partial \eta^{s,t}},$$

$$\Delta_2 = \frac{\partial \zeta(\mathbf{F}, \mathbf{Z}^s)}{\partial \mathbf{F}} \frac{\partial \delta^s \cdot \mathcal{E}^s(\eta^{s,t}, \mathbf{X}^s)}{\partial \eta^{s,t}} + \frac{\partial \zeta(\mathbf{F}, \mathbf{Z}^s)}{\partial \mathbf{Z}^s} \frac{\partial \mathcal{E}^s(\eta^{s,t}, \mathbf{X}^s)}{\partial \eta^{s,t}}. \quad (14)$$

Since $\sum_{j=1}^n \hat{g}_{ij}^s = 1$, $\hat{g}_{ij}^s > 0$, the (12) can be further rewritten as,

$$\zeta = \frac{\sum_{i,j=1}^N (\hat{g}_{ij}^s (1 - a_{ij}^s))^2}{N} + \frac{\sum_{i,j=1}^N ((1 - \hat{g}_{ij}^s) (a_{ij}^s))^2}{N^2 - N}. \quad (15)$$

Then, we can rewritten (15) into the matrix form as,

$$\zeta = \frac{\|(\mathbf{1} - \hat{\mathbf{G}}^s) \odot \mathbf{A}^s\|_F^2}{N} + \frac{\|\hat{\mathbf{G}}^s \odot (\mathbf{1} - \mathbf{A}^s)\|_F^2}{N^2 - N}, \quad (16)$$

where $\odot$ is the Hadamard product, and $\mathbf{A}^s$ represents the correlation matrix for s -th view computed from $\mathbf{C}^s$. Eq (16) measures the alignment between matrix $\hat{\mathbf{G}}^s$ and $\mathbf{A}^s$. And $\zeta(\cdot)$ achieves the minimum value when $\hat{\mathbf{G}}^s$ and $\mathbf{A}^s$ share similar distributions as well as the clustering structures.

In Fig. 3, we plot the clustering performance measured by Fscore of each view and the loss values of $\zeta(\mathbf{F}, \mathbf{Z}^s)$ with the increase of training epochs. Taking the CUB dataset as an example, the view 1 obtains better clustering performance than view 2, indicating this view contains more discriminative information than view 2. Consequently, the fused representation $\mathbf{F}$ is expected to offer clearer clustering structures compared to view-specific features $\mathbf{Z}^1$ and $\mathbf{Z}^2$. View 1 with more discriminative information will dominate the learning process. Thus, the representation $\mathbf{F}$ will contain a similar clustering structure with $\mathbf{Z}^1$, since the clustering knowledge of $\mathbf{F}$ mainly comes from $\mathbf{Z}^1$ when only employing the feature reconstruction loss. Consequently, the resultant $\hat{\mathbf{G}}^s$ and $\mathbf{G}$ both well align with $\mathbf{A}^s$, resulting a small value

for $\zeta(\mathbf{F}, \mathbf{Z}^1)$. On the contrary, $\zeta(\mathbf{F}, \mathbf{Z}^2)$ should be a large value due to the fewer contributions from $\mathbf{Z}^2$ in the learning process of the model. As a result, $\zeta(\mathbf{F}, \mathbf{Z}^1) < \zeta(\mathbf{F}, \mathbf{Z}^2)$ suggests that a more discriminative view is better at ensuring that the local similarities captured by the indicators align with the neighborhood structures of both the joint features and the original view-specific features. Consequently, view 2 must learn a clustering distribution that closely resembles the joint features to significantly reduce the loss $\zeta(\mathbf{F}, \mathbf{Z}^2)$. The clustering distribution of the joint features should also be aligned more closely with that of view 2, further facilitating the reduction of the loss $\zeta(\mathbf{F}, \mathbf{Z}^2)$. To implement this process, larger values of $\frac{\partial \zeta(\mathbf{F}, \mathbf{Z}^2)}{\partial \mathbf{F}}$ and $\frac{\partial \zeta(\mathbf{F}, \mathbf{Z}^2)}{\partial \mathbf{Z}^2}$ should be achieved, while smaller values of $\frac{\partial \zeta(\mathbf{F}, \mathbf{Z}^1)}{\partial \mathbf{F}}$ and $\frac{\partial \zeta(\mathbf{F}, \mathbf{Z}^1)}{\partial \mathbf{Z}^1}$ should be maintained. As a result, the VCR assigns a larger scale factor to the gradient of the view with less discriminative information and a lower scale factor to the gradient of the view with more discriminative information, thereby enhancing feature learning for its view-specific encoder. Ultimately, under the guidance of VCR, the model will achieve a more balanced multi-view clustering.

Thus, our proposed VCR provides additional gradients for the parameters updating in view-specific encoders and performs balanced learning for multi-view clustering.

IV. EXPERIMENTS

A. Experimental Settings

1) *Datasets*: In our experiments, ten widely used multi-view datasets, including CUB¹, HW², Youtube [68], OutdoorScene [69], RGB-D [70], nuswide [18], AWA³, xrm [18], xmedia [18], Caltech101-7 [71], are used to verify the effectiveness of the proposed method. For the nuswide, xrm, xmedia datasets, we randomly selected 3,000, 4,852, and 5,000 samples, respectively, to create the subsets nuswide-s, xrm-s, xmedia-s in our experiments. The features in all datasets are scaled into the range [0, 1] in our experiments. The detailed information of the datasets is given in Appendix 1.1. available online.

2) *Compared Methods*: To verify the superiority of proposed method, we compare it with six shallow multi-view clustering methods (e.g., AWP [32], GMC [25], LMVSC [23], EEOMVC [33], UDBGL [41], CAMVC [34]) and seven deep multi-view clustering methods (e.g., DCCA [42], MFLVC [52], DealMVC [50], CVCL [51], SCMVC [54], SCM [55], MV-CAN [15]). For all the compared approaches, we tune the hyperparameters as recommended in their respective papers to reproduce their clustering performance. The detailed hyperparameter settings for different methods are presented in Appendix 1.2. available online.

3) *Evaluation Metrics*: We employ four evaluation metrics, including accuracy (ACC), normalized mutual information (NMI), adjusted Rand index (ARI), and Fscore to measure the clustering performance of different methods. Note that a higher value of each metric indicates a better clustering result.

¹<https://www.vision.caltech.edu/visipedia/CUB-200.html>

²<https://archive.ics.uci.edu/ml/datasets/Multiple+Features>

³<https://cvml.ist.ac.at/AwA/>

TABLE I
THE CLUSTERING PERFORMANCE MEASURED BY ACC, NMI, ARI, AND FSCORE OF ALL COMPARED METHODS ON DIVERSE MULTI-VIEW DATASETS. THE HIGHEST AND THE SECOND HIGHEST VALUES UNDER EACH METRIC ARE **BOLDED** AND UNDERLINED, RESPECTIVELY.

Datasets	CUB				HW				Youtube				OutdoorScene			
Methods	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore
AWP	81.17	75.20	65.52	69.00	<u>95.80</u>	<u>91.67</u>	<u>91.11</u>	<u>91.99</u>	31.40	18.26	12.13	22.02	60.83	45.51	37.37	45.98
GMC	79.50	78.95	66.48	70.03	85.20	90.26	82.70	84.56	12.15	5.19	0.15	18.05	33.56	42.33	18.78	35.02
LMVSC	78.03	74.70	63.80	67.46	82.50	77.52	70.87	73.85	25.01	10.06	5.33	14.97	54.67	39.20	31.11	39.89
EEOMVC	65.00	66.49	51.32	56.43	70.15	63.81	55.79	60.35	26.25	11.80	7.08	17.27	57.70	40.35	29.64	39.02
UDBGL	79.35	<u>79.82</u>	68.91	72.17	92.14	85.62	83.69	85.32	38.03	22.74	16.39	24.84	69.46	52.80	45.79	52.77
CAMVC	81.66	79.67	69.65	72.73	91.24	85.14	82.31	84.08	25.70	11.58	6.60	16.13	71.16	55.82	48.31	54.91
DCCA	60.09	55.87	40.79	48.84	85.04	85.15	77.75	83.50	10.68	1.48	0.01	19.07	48.16	51.58	36.42	49.76
MFLVC	67.00	62.91	49.24	55.59	82.65	80.41	74.02	77.14	39.70	28.84	20.02	29.36	64.62	55.15	42.90	51.11
DealMVC	61.00	66.10	51.27	58.04	82.10	78.73	72.57	75.42	36.60	23.98	15.88	26.24	75.26	62.73	54.21	61.83
CVCL	79.33	71.03	61.54	66.17	92.95	87.53	85.43	87.15	<u>46.15</u>	<u>31.68</u>	<u>23.32</u>	<u>31.50</u>	71.65	60.23	50.34	57.98
SCMVC	72.83	68.71	55.82	61.66	90.95	84.76	81.43	83.85	41.60	29.72	20.27	29.08	72.10	60.57	52.56	59.89
SCM	<u>84.83</u>	79.11	<u>71.23</u>	<u>75.17</u>	89.25	80.57	77.63	80.72	37.20	23.01	16.45	24.73	60.45	51.95	41.59	49.93
MVCAN	80.67	78.42	67.77	72.99	95.75	91.58	90.95	91.98	29.75	22.31	13.64	25.35	<u>75.89</u>	<u>63.17</u>	<u>56.01</u>	<u>62.50</u>
Ours	86.67	82.21	75.53	78.44	97.95	95.18	95.46	96.01	47.67	34.25	26.23	34.70	78.46	64.46	58.64	65.12
Datasets	RGB-D				nuswide-s				xrmb-s				xmedia-s			
Methods	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore
AWP	41.34	28.95	18.55	27.03	55.93	47.05	36.38	44.37	24.07	38.19	13.66	16.26	81.12	84.86	74.32	74.95
GMC	42.24	30.97	8.81	29.88	22.83	19.60	1.72	19.68	13.09	23.20	0.41	6.53	80.66	89.34	54.23	55.96
LMVSC	<u>45.50</u>	<u>39.23</u>	24.94	32.83	57.02	42.89	34.93	41.65	20.89	34.13	10.75	13.34	78.45	90.57	76.47	77.09
EEOMVC	43.55	36.70	22.85	32.18	61.27	45.02	39.80	46.09	26.88	39.16	14.79	17.27	<u>89.20</u>	92.53	<u>87.61</u>	<u>87.93</u>
UDBGL	43.69	38.36	<u>25.71</u>	33.32	64.49	52.38	45.59	51.25	29.36	44.05	17.63	20.08	85.74	<u>93.80</u>	86.63	86.98
CAMVC	35.14	29.08	16.89	25.33	68.05	54.88	51.19	56.20	29.00	43.86	17.93	20.35	85.36	92.47	85.88	86.24
DCCA	17.19	6.51	2.32	13.58	48.16	40.03	22.89	41.40	23.15	37.09	10.88	19.42	49.68	54.96	31.46	40.99
MFLVC	38.03	21.41	18.37	29.24	32.10	20.75	16.09	24.40	24.69	32.82	13.17	17.52	13.28	26.71	6.93	12.33
DealMVC	45.27	27.05	23.14	<u>37.64</u>	51.93	37.88	32.75	38.81	17.19	27.59	8.83	14.61	10.20	22.72	5.86	10.10
CVCL	25.86	17.05	8.02	21.88	58.63	48.30	39.92	47.41	27.10	38.92	14.44	18.32	34.02	51.15	24.53	30.46
SCMVC	31.06	23.25	15.50	25.28	53.97	40.63	34.91	41.37	29.78	43.76	18.02	21.84	33.18	48.07	26.32	26.09
SCM	35.20	30.35	17.84	29.31	57.63	44.17	38.51	44.56	29.55	41.92	16.52	20.14	85.96	90.71	83.96	84.19
MVCAN	40.99	33.71	22.33	32.82	62.93	49.58	44.32	49.86	32.11	<u>45.99</u>	<u>18.95</u>	<u>23.05</u>	86.46	93.13	86.65	87.43
Ours	45.84	39.66	27.34	38.03	<u>66.49</u>	<u>53.94</u>	<u>49.53</u>	<u>54.03</u>	<u>31.03</u>	46.56	19.33	23.22	90.43	94.48	89.20	90.21
Datasets	nuswide				AWA				xmedia				xrmb			
Methods	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore
AWP	41.82	39.98	22.13	33.44	63.61	70.85	52.91	54.03	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
GMC	10.82	2.56	0.02	18.14	13.65	24.04	0.70	5.19	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
LMVSC	60.64	44.36	39.49	45.66	29.51	36.35	15.84	17.74	77.95	<u>91.57</u>	74.63	74.77	19.33	28.25	9.88	12.76
EEOMVC	57.01	41.65	34.54	41.28	61.99	68.91	52.08	53.17	N/A	N/A	N/A	N/A	24.99	34.72	14.36	17.06
UDBGL	61.42	47.64	41.90	47.73	<u>66.94</u>	76.29	<u>58.67</u>	<u>59.61</u>	46.49	61.96	23.75	24.24	24.25	34.12	13.45	16.34
CAMVC	<u>63.93</u>	<u>49.80</u>	<u>44.19</u>	<u>49.83</u>	64.43	73.44	56.33	57.31	<u>83.12</u>	93.28	80.72	<u>80.82</u>	27.35	39.09	16.65	19.23
DCCA	43.35	35.50	17.33	37.21	42.50	55.31	26.86	34.88	8.51	26.56	2.62	5.62	20.88	29.48	10.50	16.82
MFLVC	35.82	27.06	18.94	28.11	27.52	40.17	18.28	20.25	N/A	N/A	N/A	N/A	26.91	38.05	16.01	19.82
DealMVC	56.47	41.25	35.69	42.47	38.78	50.35	28.23	28.67	10.20	22.72	5.86	10.10	25.17	35.18	15.83	19.85
CVCL	57.99	43.22	38.22	44.13	35.76	49.15	23.46	26.13	12.87	42.52	7.73	9.01	22.31	36.59	13.26	16.93
SCMVC	48.71	37.40	26.62	36.99	38.15	51.35	28.46	28.65	9.28	36.05	5.72	6.38	27.11	41.07	18.00	<u>20.67</u>
SCM	53.07	39.53	31.43	38.92	48.42	57.71	36.47	36.85	62.80	73.93	49.84	50.18	28.31	38.09	16.55	19.53
MVCAN	59.12	44.30	38.12	44.56	56.43	68.24	47.68	48.79	73.27	86.83	66.87	70.50	<u>29.08</u>	39.93	17.31	20.49
Ours	68.68	54.29	50.18	55.08	<u>67.16</u>	<u>75.97</u>	59.17	60.89	85.44	91.00	<u>79.73</u>	81.48	29.20	<u>39.97</u>	<u>17.80</u>	20.81

4) *Implementation Details*: In our proposed method, the view-specific encoders and decoders are all implemented in the fully connected architectures. The detailed structures of the encoders and decoders are D -196-128-64 and 64-128-196- D , respectively, where D is the input and output dimensions of encoders and decoders. The proposed method is implemented via the PyTorch [72] tool and conducts experiments on a single NVIDIA 2080Ti GPU with a Ubuntu 20.04 platform. The Adam optimizer [73] is utilized to optimize the proposed method, and the initial learning rate is set to 0.001. The model is trained for 3000 epochs on all datasets, and the obtained joint feature representation is used to obtain the clustering results via K-means. Besides, the trade-off parameter λ is searched in range $[10^{-5}, 10^{-4}, \dots, 10^4, 10^5]$ with a grid search scheme.

B. Clustering Performance Evaluation

The clustering results measured by four metrics on eight multi-view datasets of all compared methods are reported in Table I. From this table, we obtain the following observations: First, the proposed BMvC method consistently performs better than other compared methods on most datasets. For instance, BMvC achieves 78.46, 64.46, 58.64, and 65.12 percentages in terms of ACC, NMI, ARI, and Fscore on the OutdoorScene dataset. Such results exceed the second performer (MVCAN) by about 2.57, 1.29, 2.63, and 2.62 percentages measured by ACC, NMI, ARI, and Fscore on the OutdoorScene dataset. Such clustering performance benefits strongly demonstrate the effectiveness and superiority of the proposed BMvC method, which facilitates more balanced multi-view learning through the introduction of view-specific contrastive regularization. Second, the

proposed BMvC method is superior to other deep-based multi-view clustering competitors, which seldom take the imbalanced learning problem in multi-view clustering into consideration. The MVCAN method assigns different views with weights based on the mutual information between joint clustering indicators and view-specific ones to mitigate the side effects of less robust views. However, this approach may further exacerbate the issues of imbalanced learning, as the information in certain views with lower weight allocations is more difficult to learn. Third, compared with other shallow multi-view clustering methods, our BMvC method achieves superior clustering results. We note that the graph-based methods, e.g., AWP, GMC, and UDBGL, usually obtain considerable performance. This indicates that effectively leveraging the similarities among different samples is beneficial for clustering tasks.

To verify the effectiveness of our proposed method for large-scale multi-view clustering tasks, we conduct experiments on the nuswide, AWA, xmedia, and xrmc datasets. In Table I, we present the clustering performance of various methods. The results indicate that our proposed method consistently outperforms competitors, demonstrating its effectiveness in handling large-scale multi-view datasets.

In addition to the external metrics, we introduce six internal metrics to evaluate the clustering performance of different methods, with the experimental results and analysis presented in Appendix 1.3. available online. To investigate the impact of varying the number of views on the clustering performance of the proposed method, we conduct experiments on the Caltech101-7 dataset as the number of views increases, and the results and analysis can be found in Appendix 1.4. available online. We also assess the effectiveness of the proposed method in handling noisy data, with results and analysis detailed in Appendix 1.5. available online. In Appendix 1.6, available online we analyze the computational complexity of the proposed method and provide a clearer comparison of the training times among various methods, including our approach on the nuswide dataset.

C. Ablation Studies

In this section, we conduct a series of experiments on the eight multi-view datasets to analyze the effectiveness of diverse components in the proposed method.

1) *Effectiveness of View-Specific Contrastive Regularization:* The view-specific contrastive regularization, which leverages the similarities captured from original features and joint features to perform contrastive learning on the view-specific clustering space, ensuring additional gradients for the parameters updating in each view-specific encoder, is the key component of the proposed method. To verify its effectiveness, we formulate three methods here, i.e., Rec, VCR, and Rec + VCR. For the Rec method, the view-specific contrastive regularization is removed from the BMvC method and only the feature reconstruction loss is utilized to optimize the model. VCR method cuts out the view-specific decoder parts and just leverages the view-specific contrastive regularization loss to optimize the model. Rec+VCR is set as the same as our proposed BMvC methods, in which the feature reconstruction and view-specific contrastive

regularization are jointly exploited. The clustering performance in terms of ACC, NMI, ARI, and Fscore on eight multi-view datasets of the above formulated methods is reported in Table II. From the results, we find that 1) The VCR method performs better than Rec in most cases, which indicates the VCR part can obtain more balanced multi-view learning during training, ensuring that the complementary information of multi-view datasets can be more explored. 2) Rec + VCR is consistently superior to Rec and VCR, demonstrating that jointly utilizing Rec and VCR is the best choice for multi-view clustering.

To verify the proposed VCR is the ideal choice for balanced multi-view clustering task, we introduce two methods, e.g., the graph Laplacian regularization (GLR) loss and soft neighbors-based contrastive learning (SCL) loss [74], which are two terms most similar to the proposed VCR. The mathematical formulations of GLR and SCL losses tolerated the balanced multi-view clustering task are,

$$\zeta_g = \sum_{i,j=1}^N \hat{g}_{ij}^s \left\| \frac{\mathbf{c}_i^s}{\|\mathbf{c}_i^s\|} - \frac{\mathbf{c}_j^s}{\|\mathbf{c}_j^s\|} \right\|_2^2 = 2 \sum_{i,j=1}^N \hat{g}_{ij}^s (1 - a_{ij}^s), \quad (17)$$

$$\zeta_s = \frac{1}{N} \sum_{i=1}^N -\log \left(\frac{\sum_{j=1}^N \hat{g}_{ij}^s e^{a_{ij}^s}}{\sum_{r=1}^N e^{a_{ir}^s}} \right), \quad (18)$$

where ζ_g and ζ_s denote GLR loss and SCL loss. Compared with (12), (17) and (18), the proposed VCR has the following advantages: 1) The squared error terms in (12) penalize large mistakes more strongly than Eqs. (12), and (18), leading to more adjusting in learning procedure for those samples in weak views. 2) (12) has a symmetrical formulation for positive and negative sets, making samples in positive and negative sets can be all constrained. To verify the effectiveness of the proposed VCR, we replace VCR with the GLR and SCL and conduct experiments, from which the experimental results are reported in Table II. The proposed VCR performs better than GLR and SCL on most of the datasets, verifying the superiority of the proposed VCR for handling the unbalanced multi-view clustering issue. The results of the proposed method with GLR are sometimes worse than those of the proposed method with only the feature reconstruction loss, indicating that only regularizing the data samples in the positive set may not enforce the data with structural information. The proposed method with SCL outperforms that that only with Rec, revealing SCL is effective but not the best choice for dealing with the unbalanced multi-view clustering issue.

2) *Impacts of Different Feature Fusion Manners:* To study the impacts of different multi-view feature fusion manners on the final multi-view clustering model, we construct three methods, i.e., BMvC-w-Asum, BMvC-w-Wsum, and BMvC-w-Cat, which fuse multi-view features via average feature addition, feature addition with sample-wise weights, and feature concatenation, respectively. As can be seen from the results in Table II, our BMvC method equipped with average feature addition, feature addition with sample-wise weights, and feature concatenation can both obtain considerable clustering performance. This indicates that our proposed BMvC method is less sensitive to different multi-view feature fusion manners. The proposed method with feature concatenation fusion manner performs

TABLE II
THE ABLATION STUDY OF THE PROPOSED METHOD IN TERMS OF ACC, NMI, ARI, AND FSCORE ON EIGHT BENCHMARK MULTI-VIEW DATASETS

Datasets	CUB				HW				Youtube				OutdoorScene			
Methods	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore
(a) Effectiveness of View-specific Contrastive Regularization																
Rec	78.54	77.43	65.44	71.99	90.77	83.89	81.40	83.55	27.48	21.29	10.93	24.80	72.06	56.98	49.12	56.95
VCR	83.81	80.70	72.86	76.10	97.34	94.25	94.13	94.87	42.91	30.84	22.81	32.15	73.49	60.08	52.36	59.33
Rec + GLR	75.12	78.42	64.21	72.65	93.65	89.13	86.24	88.62	25.90	18.94	9.23	23.11	65.81	56.20	44.42	54.24
Rec + SCL	81.37	80.47	70.30	75.80	93.18	89.86	85.43	88.79	38.26	29.50	17.48	29.62	73.43	60.65	51.58	59.48
Rec + VCR	86.67	82.21	75.53	78.44	97.95	95.18	95.46	96.01	47.67	34.25	26.23	34.70	78.46	64.46	58.64	65.12
(b) Impacts of Diverse Feature Fusion manner																
CMvC-w-ASum	89.30	85.57	80.19	82.49	97.47	94.41	94.39	95.13	47.01	32.71	25.14	33.43	75.87	62.23	55.77	62.15
CMvC-w-WSum	88.32	84.90	79.07	81.52	97.51	94.62	94.50	95.22	46.14	32.09	24.88	32.91	78.34	64.01	58.49	64.75
Ours-w-Cat	86.67	82.21	75.53	78.44	97.95	95.18	95.46	96.01	47.67	34.25	26.23	34.70	78.46	64.46	58.64	65.12
Datasets	RGB-D				nuswide-s				xrm-b-s				xmedia-s			
Methods	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore	ACC	NMI	ARI	Fscore
(a) Effectiveness of View-specific Contrastive Regularization																
Rec	38.34	33.91	20.99	31.67	59.63	47.69	39.75	46.95	29.30	44.17	17.25	21.45	85.21	93.39	85.94	87.35
VCR	43.01	40.61	26.07	37.50	68.06	54.49	50.78	55.16	28.19	44.99	17.71	21.70	87.74	94.75	88.40	89.69
Rec + GLR	20.32	9.37	2.62	17.28	N/A	N/A	N/A	N/A	31.62	46.25	19.84	23.78	90.41	96.71	90.34	91.07
Rec + SCL	42.17	40.08	23.29	36.83	64.23	53.10	45.79	52.79	30.22	44.21	17.69	21.86	89.95	95.11	89.07	90.51
Rec + VCR	45.84	39.66	27.34	38.03	66.49	53.94	49.53	54.03	31.03	46.56	19.33	23.22	90.43	94.48	89.20	90.21
(b) Impacts of Diverse Feature Fusion manner																
CMvC-w-ASum	45.28	38.98	26.85	37.87	67.44	54.19	49.91	54.61	30.79	46.32	19.17	23.26	90.97	95.17	90.16	91.17
CMvC-w-WSum	44.37	41.98	27.37	39.03	66.57	53.91	48.21	54.08	30.71	45.74	18.86	22.78	87.15	94.24	87.91	88.99
Ours-w-Cat	45.84	39.66	27.34	38.03	66.49	53.94	49.53	54.03	31.03	46.56	19.33	23.22	90.43	94.48	89.20	90.21

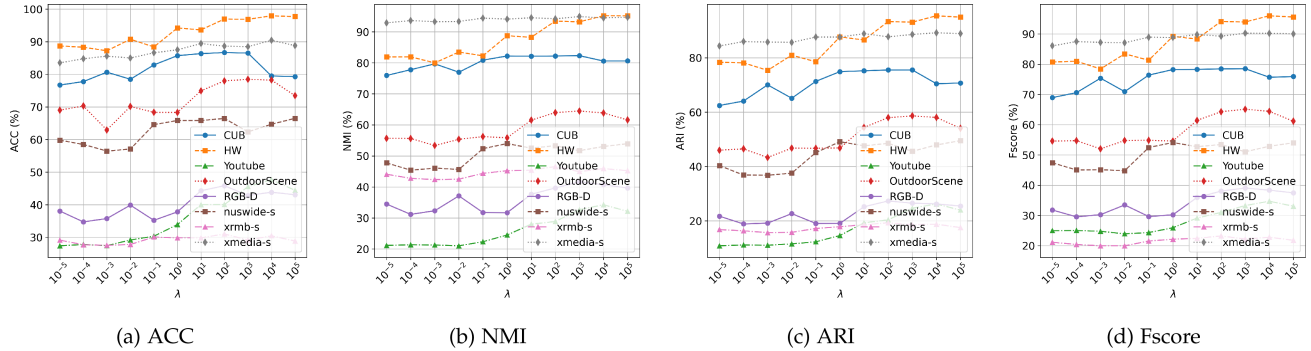


Fig. 4. The parameter sensitivity of the proposed method on eight benchmark multi-view datasets in terms of ACC, NMI, ARI, and Fscore, respectively.

better than other manners on more datasets and metrics. In this paper, we choose the feature concatenation as the default since this manner is commonly used in multi-view feature fusion and does not introduce heavy computational affords compared with others.

3) *Parameter Sensitivity Analysis*: Our proposed BMvC method consists of a key balance parameter λ to trade-off the feature reconstruction and view-specific contrastive regularization loss. To study the parameter sensitivity, we give the clustering performance measured by ACC, NMI, ARI, and Fscore on eight multi-view datasets varying with different λ in Fig. 4. From the results, we observe that the clustering performance of the BMvC method slightly fluctuates with the λ . Additionally, when the parameter λ is in the range $[10^0, 10^1, 10^2, 10^3]$, our BMvC method can obtain considerable clustering results in most datasets. Therefore, we suggest setting the parameter λ in the range $[10^0, 10^1, 10^2, 10^3]$ when it is applied for some applications.

Apart from the hyper-parameter λ , the number of nearest neighbors will affect the effectiveness of the view-specific contrastive regularization. To evaluate the impacts of different number of nearest neighbors, we vary the number of nearest neighbors over $[5, 10, 15, 20, 25, 30, 35, 40]$ on eight multi-view datasets and report the results in Fig. 5. As shown, the proposed method is relatively insensitive to the choice of nearest neighbors and consistently achieves strong clustering performance for values greater than 5.

D. Application for Spatially Resolved Transcriptomics Data Clustering

The spatial arrangement of various cell types is closely linked to the functionality of complex tissues. Recent advancements in spatial transcriptomics (ST) have transformed our capacity to investigate this relationship. By combining transcriptomic data with their respective spatial contexts, researchers can achieve

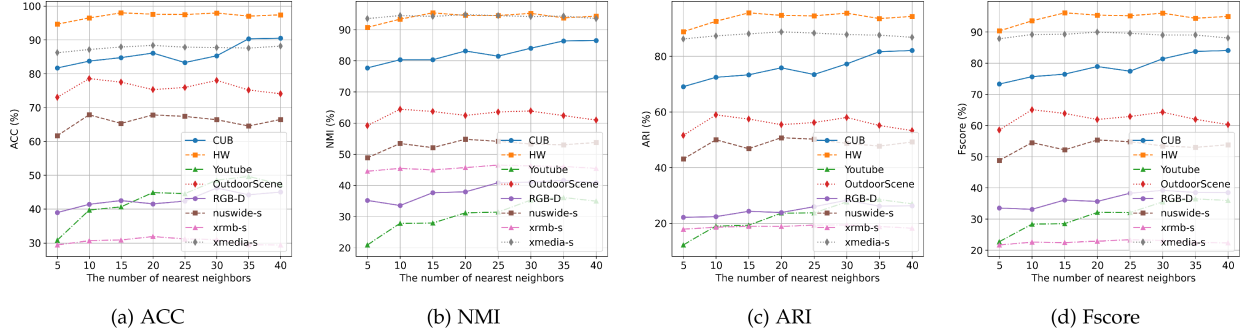


Fig. 5. The effects of varying the number of neighbors on the performance of the proposed method across eight benchmark multi-view datasets, evaluated using ACC, NMI, ARI, and F-score metrics.

a more profound understanding of the biological roles of these tissues. Furthermore, spatial information aids in mapping inter-cellular communication, thereby deepening our comprehension of cell interactions and the regulatory mechanisms that influence cell behavior. The fusion of spatial and transcriptomic data provides a more holistic view of complex tissue functions.

The graph convolutional network (GCN) is a useful tool to encode the information from both gene expression and spatial location. Based on GCN, various methods are proposed for spatial transcriptomics data clustering [75], [76], [77], [78]. For example, Xu et al. [79] integrated the morphological information into gene expression to construct extended features, which are fed into the GCN for low-dimensional feature representation learning. Differently, Hu et al. [80] constructed a similarity graph by jointly utilizing the information of spatial location and histology image for GCN-based feature learning. Wang et al. [81] presented triple GCNs that extract unique information from gene expressions, spatial neighbors, and their shared information. Recently, several methods for clustering spatial transcriptomics data have been proposed based on self-supervised learning. For instance, Zhu et al. [82] introduced a contrastive learning-based dual reduction loss aimed at minimizing the correlation between features derived from two different views, thereby enhancing feature representation for clustering. He et al. [83] designed a heterogeneous graph guided contrastive learning framework to jointly capture the information from both genes and spatial locations. In addition to contrastive learning, various masked mechanisms in self-supervised learning have also been employed [84], [85], [86]. A notable example is SpaMask [86]. This approach utilizes masked graph autoencoders to improve clustering accuracy through node masking, while masked graph contrastive learning enhances the embeddings of adjacent nodes by leveraging spatial proximity and feature similarity. Wen et al. [87] designed a single-cell pre-trained transformer that leverages spatially-resolved transcriptomic data in pre-training to facilitate learning cell-cell relationships. The heterogeneity between gene expression and spot spatial location results in an imbalanced multi-view learning issue in previous spatial transcriptomics clustering methods. For example, the spatial continuity of cellular tissue makes it preferable to learn the information from the spatial location rather than the gene expression for clustering. Both the gene expression and spot spatial location information are important for revealing the

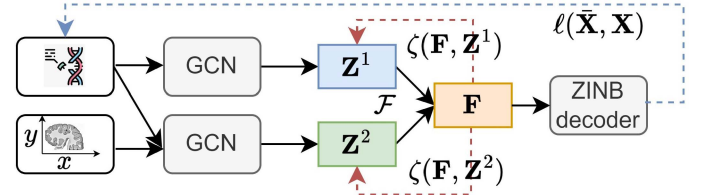


Fig. 6. Illustration of the proposed BMvC method on the spatially resolved transcriptomics data clustering task.

cellular relationships. Thus, it is necessary to formulate a balanced multi-view learning framework for the spatially resolved transcriptomics data clustering task.

In this section, we transfer our proposed BMvC method to deal with the spatially resolved transcriptomics data clustering task, as shown in Fig. 6. Specifically, two graph convolutional neural networks are employed to extract the transcriptomic expression-related representation and spatial distribution-related representation to comprehensively depict each cell. Then, the feature concatenation operator with a linear layer is utilized to fusion such two different types of information. As done in previous methods [82], [83], the ZINB decoder [88], which captures the complex global information of data, is introduced to reconstruct the gene expression. Finally, the view-specific contrastive regularization is leveraged to achieve more balanced multi-view learning in the spatially resolved transcriptomics data clustering task. Thus, the total loss functions used in the proposed method for spatially resolved transcriptomics data clustering are,

$$\mathcal{L} = \ell(\hat{\mathbf{X}}, \mathbf{X}) + \sum_{r=1}^2 \lambda \zeta(\mathbf{F}, \mathbf{Z}^r), \quad (19)$$

where $\mathbf{X}$ and $\hat{\mathbf{X}}$ denote the gene matrix and reconstructed variables, respectively. $\mathbf{Z}^1$ and $\mathbf{Z}^2$ represent the low-dimensional feature representations obtained from two GCN feature extractors. $\mathbf{F}$ is the fused feature representation. $\zeta(\mathbf{F}, \mathbf{Z}^r)$ represents the proposed view-specific contrastive loss. The reconstruction loss $\ell(\hat{\mathbf{X}}, \mathbf{X})$ is implemented using the negative log-likelihood of the ZINB distribution [88]. Let $\hat{\mathbf{X}}$ consist of three variables, e.g., dropout rate π , dispersion degree μ , and mean θ , all estimated by the ZINB decoder from the fused feature representation $\mathbf{F}$. The

TABLE III

THE SPATIAL CLUSTERING PERFORMANCE MEASURED BY NMI AND ARI OF ALL COMPARED METHODS ON TWO SPATIALLY RESOLVED TRANSCRIPTOMICS DATASETS. THE HIGHEST AND THE SECOND HIGHEST VALUES UNDER EACH METRIC ARE **BOLDED** AND UNDERLINED, RESPECTIVELY.

Datasets	HBC		MBA	
Methods	NMI	ARI	NMI	ARI
K-means	51.45	35.12	58.75	24.45
SCANPY	51.59	49.15	45.29	24.43
SpaGCN	64.34	56.12	62.72	34.23
DeepST	68.75	53.13	56.71	25.63
SCGDL	43.81	35.32	64.30	26.95
GraphST	67.00	52.65	68.89	38.59
Spatial-MGCN	69.81	64.32	71.13	42.46
stLearn	62.64	55.26	65.52	38.42
CellPLM	33.13	19.12	57.17	24.77
stMMR	<u>69.65</u>	<u>65.65</u>	68.39	<u>48.04</u>
MAFN	69.32	61.88	72.76	43.58
stGCL	69.27	65.62	69.25	47.63
Ours	68.27	67.01	<u>72.70</u>	50.81

ZINB-based feature reconstruction loss is defined as follows,

$$\begin{aligned}
 \ell(\hat{\mathbf{X}}, \mathbf{X}) &= -\log(\text{ZINB}(\mathbf{X}|\pi, \mu, \theta)) \\
 &= \pi \cdot \delta_0(\mathbf{X}) + (1 - \pi) \cdot \frac{\Gamma(\mathbf{X} + \theta)}{\mathbf{X}! \Gamma(\theta)} \\
 &\quad \left(\frac{\theta}{\theta + \mu} \right)^\theta \left(\frac{\mu}{\theta + \mu} \right)^{\mathbf{X}}, \quad (20)
 \end{aligned}$$

where $\delta_0(\cdot)$ denotes an indicator function that equals 1 when $\mathbf{X} = 0$, and 0 otherwise. $\Gamma(\cdot)$ represents the Gamma function.

1) *Datasets*: We evaluate the proposed method on two spatially resolved transcriptomics datasets:

Human Breast Cancer (**HBC**) [89]: This dataset features 20 meticulously annotated regions categorized into four main morphological types.

Mouse Brain Anterior Tissue (**MBA**) [90]: This dataset is labeled with 52 distinct class areas and captured using the 10X Visium device, and shows minimal inter-class variation.

Following previous methods [82], [83], we utilized the SCANPY toolkit [91] to filter out genes exhibiting minimal expression levels or variance. We then selected the top 3000 genes with the highest variability for the subsequent clustering task. Finally, the expression levels of these genes were normalized using a scaling factor.

2) *Compared Methods*: To verify the superiority of proposed method, we compare it with twelve benchmark methods: K-means, SCANPY [91], SpaGCN [80], DeepST [79], SCGDL [75], GraphST [76], Spatial-MGCN [81], stLearn [77], CellPLM [87], stMMR [78], MAFN [82], and stGCL [83]. In our experiments, we set the parameters of all compared methods as suggestions according to their papers to implement their best clustering performance.

3) *Spatial Clustering Performance*: We report the spatial clustering performance in terms of ARI and NMI of all compared methods in Table III. From the results, we note that our BMvC method achieves superior results compared with previous well-designed spatial clustering methods on the two datasets. For example, our BMvC method obtains the best performance in terms

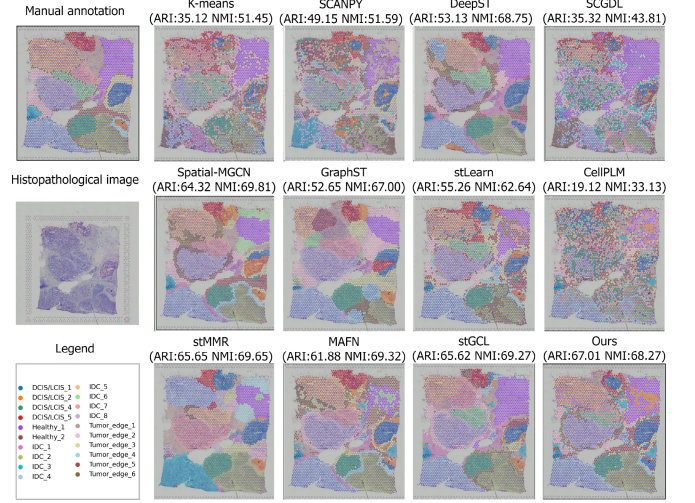


Fig. 7. Clustering distribution visualization results of compared methods on HBC dataset.

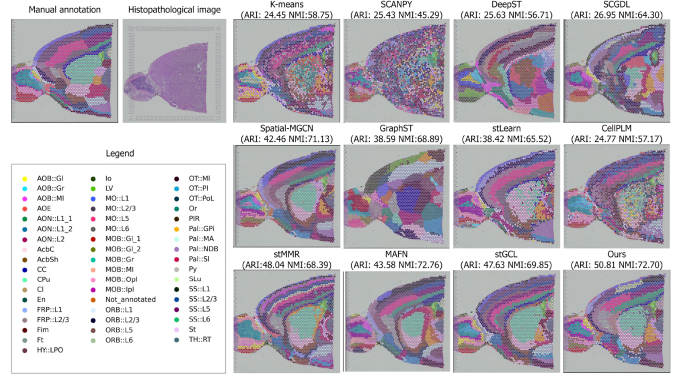


Fig. 8. Clustering distribution visualization results of compared methods on MBA dataset.

of ARI on HBC and MBA datasets, and is better than the second performer by about 1.36 and 2.81 percentages. Previous spatial clustering methods usually follow the joint-training paradigm to perform multi-view learning. As mentioned in Section II-B2, the joint-training paradigm poses an imbalanced learning problem, making the view-specific encoders under-optimized. Thus, such joint-training paradigm-based methods have limited spatial clustering performance. In contrast, our BMvC method introduces a view-specific contrastive regularization to find a better trade-off between the exploitation of view-specific patterns and the exploration of view-invariance patterns to fully learn the multi-view information for the clustering task.

4) *Visualization Analysis*: The clustering distribution visualization results of all compared methods are presented in Figs. 7 and 8. Compared with other methods, our method provides more compact and region-consistent visualization results, especially in the left bottom corner and left part of the HBC and MBA datasets, respectively. Furthermore, the proposed method obtains clearer spatial boundaries, indicating more accurate cell clustering. Such visualization superiority of the proposed

method remarkably demonstrates the effectiveness of the proposed method and the efficiency of the well-designed view-specific contrastive regularization in dealing with the imbalanced multi-view problem.

V. CONCLUSION

In this paper, we proposed a novel balanced multi-view clustering method to achieve more balanced multi-view learning and further improve the clustering performance. We first analyzed the imbalanced multi-view clustering problem existing in the joint-training paradigm from the gradient view. Then, we introduced a view-specific contrastive regularization to make a better trade-off between the exploitation of view-specific patterns and the exploration of view-invariance patterns to fully learn the multi-view information for the clustering task. Additionally, the theoretical analysis was provided to verify the effectiveness of the VCR from the gradient perspective. Finally, Extensive experiments on various benchmark multi-view clustering datasets and spatially resolved transcriptomics datasets were conducted to verify the efficacy of our method.

The proposed method assumes that multi-view data is fully observed and that instances across all views are well aligned. However, in real-world scenarios, information incompleteness is often present in multi-view data due to varying working conditions. Therefore, it is essential to extend the proposed method to address this issue of information incompleteness. In the future, we will design effective methods based on cross-view sample prediction and cross-view correspondence prediction to address the challenges posed by partial multi-view data and partially view-aligned multi-view data. Additionally, the computational complexity of the proposed method is given by $\mathcal{O}(T \cdot (\sum_{r=1}^M D^r \cdot N_b^2 + M \cdot N_b \cdot c^2) \cdot \lceil \frac{N}{N_{mb}} \rceil)$, which primarily depends on the batch size, the number of samples, and the total number of training epochs. In our future work, we also aim to develop a more efficient approach, e.g., the coreset-based method, for the proposed method to handle large-scale data.

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Zhenglai Li (Member, IEEE) received the BE, MS, and PhD degrees from the China University of Geosciences, Wuhan, China, in 2018, 2021, and 2024, respectively. He is currently a postdoctoral research fellow with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China. He has authored or coauthored 20 peer-reviewed papers in highly regarded journals and conferences, including *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Image Processing*,

IEEE Transactions on Circuits and Systems for Video Technology, *IEEE Transactions on Geoscience and Remote Sensing*, and ACM MM. He was on the Technical Program Committees for leading conferences, such as NeurIPS, ICLR, KDD, and ACM MM. His current research interests include machine learning and computer vision, with a focus on unsupervised learning, clustering, multi-view/multi-modal representation learning algorithms, and their applications in healthcare.



JunWang received the MS degree from the China University of Geosciences, Wuhan, China, in 2023. He is currently working toward the PhD degree with the School of Computer Science, National University of Defense Technology. His research interest focuses on multi-view learning.



Chang Tang (Senior Member, IEEE) received the PhD degree from Tianjin University, Tianjin, China, in 2016. He joined the AMRL Lab with the University of Wollongong from 2014 to 2015. He is currently an associate professor with the School of Software Engineering, Huazhong University of Science and Technology, Wuhan, China. He was with the School of Computer Science, China University of Geosciences, Wuhan, China, from 2017 to 2025. He has authored or coauthored more than 100 peer-reviewed papers, including those in highly regarded journals and conferences such as *IEEE Transactions on Pattern Analysis and Machine Intelligence*,

IEEE Transactions on Multimedia, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Human-Machine Systems*, ICCV, CVPR, IJCAI, AAAI and ACM MM. His current research interests include multi-view learning and pattern recognition. He was an associate editor of *Neural Networks*, *IEEE Journal of Biomedical and Health Informatics*, *CAAI Transactions on Intelligence Technology*, and *BMC Bioinformatics*. He was on the Technical Program Committees or as Area Chair of some top conferences, such as NIPS, ICML, CVPR, ICCV, ECCV, IJCAI, ICME, and AAAI.



general object detection) and deep learning, and their applications. He is a member of ACM.

Xinzhong Zhu (Member, IEEE) received the PhD degree from Xidian University, Xi'an, China, in 2018. He is currently a professor with the College of Mathematics, Physics and Information Engineering, Zhejiang Normal University, Jinhua, China. His research interests include machine learning, computer vision, manufacturing informatization, robotics and system integration, and intelligent manufacturing, and focusing on kernel learning and feature selection, multi-view clustering algorithms, real-time object detection (e.g., pedestrian detection, vehicle detection, and general object detection) and deep learning, and their applications. He is a member of ACM.



Wei Zhang received the BE degree from Zhejiang University in 2004, the MS degree from Liaoning University in 2008, and the PhD degree from the Shandong University of Science and Technology in 2018. He is currently an associate professor with the Shandong Computer Science Center (National Supercomputing Center in Jinan), Qilu University of Technology (Shandong Academy of Sciences). His research interests include future generation network architectures, edge computing, and edge intelligence.



Xinwang Liu (Senior Member, IEEE) received the PhD degree from the National University of Defense Technology (NUDT), China. He is currently a full professor with the School of Computer, NUDT. He has authored or coauthored more than 60 peer-reviewed papers, including those in highly regarded journals and conferences, such as *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Image Processing*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Information Forensics and Security*, NeurIPS, ICCV, CVPR, AAAI, IJCAI. His current research interests include kernel learning and unsupervised feature learning. More information can be found at xinwangliu.github.io.